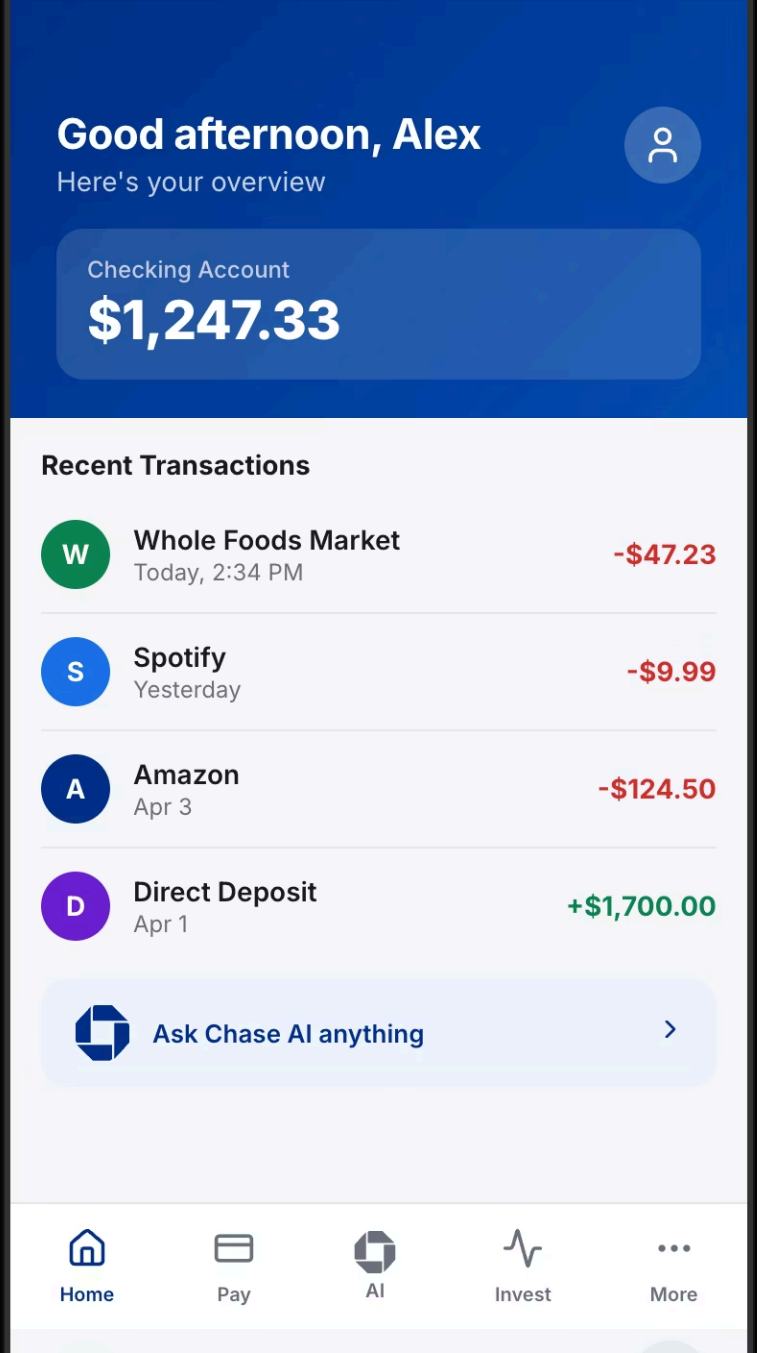
Agentic Banking for Chase:

Vincent Park · Staff PM Case Study



About Me

Vincent Park

Career Arc

Engineer → PM → Founder. A builder at heart who has shipped AI-powered products at every stage of the company lifecycle.

Relevant AI/ML Experience

- Dropbox — Growth Product Lead with focus on ML-led user activation, onboarding, and trials
- Persona — Product Lead on Document AI, powering millions of documents processed weekly for Fortune 10 partners
- Scoop — agentic workflows for biotech and pharma regulatory submissions
- Deep interest in safety-pilled Agentic AI products

PROJECT #1

Persona Document AI — ML-to-LLM Migration

The Problem

Each new document type required **1-2 weeks** of training data and model development work. With ~10-20 types across each country, the requirement gap kept widening — so we needed eval-first rigor and phased migration in a regulated domain.

4x

YoY Revenue in 2024 and 2025

150+

Document Types supported

Found the Inflection Point

Ran 50 financial docs through GPT-3.5 in one evening. The insight: LLMs improve with every release, while ML had hit its ceiling.

Designed a Parallel Migration

Built a 500+ doc eval harness tracking accuracy, pass rate, latency, and cost. Each document type had clear go/no-go gates.

Drove GTM Packaging

Created KYB, insurance, and fraud verticals with tailored decks, demos, and ROI metrics.

PROJECT #2

Dropbox Trials Platform — Data-Driven Growth System

The Problem

Dropbox's free-to-paid conversion system was hardcoded, so every experiment required engineering work. Trials were one-size-fits-all, and simple A/B tests took weeks.

$8M+

Incremental ARR

20+

Trial experiments per Quarter

Built the Platform

Designed a configurable rules engine so PMs could change trial settings **without code**.

Unlocked the Insight

Analyzed 18 months of trial data and found a non-linear link between trial length and conversion.

Personalized at Scale

DS built a conversion propensity model that personalized trial length at the individual level.

Case Study

The Business Case

The Bank's Strategic Imperative

Revenue Engines

- Deposits → lending base
- Interchange fees on card spend
- Product cross-sell & reduced churn

The White Space

Digital availability ≠ digital intelligence. Every bank has an app that **shows you your data.** No bank has an agent that **acts on it.**

The opportunity: turn the banking app from a place you *check* into a place that **works for you.**

Customer Segmentation

Three Generations, One Platform

Segment	Profile	AI Opportunity
Boomers (57–75)	Highest deposits, strong loyalty, growing digital adoption	Wealth preservation, fraud alerts, retirement planning
Millennials (29–44)	Peak earnings, managing mortgages + debt, financially stretched	Debt optimization, goal-based savings, family planning
Gen Z (18–28)	Fastest-growing by new accounts, low deposits today, highest lifetime value	**In financial crisis — and the most receptive to AI**

Why Gen Z is the strategic beachhead:

Gen Z: Highest Defection Risk

Chime captured **13%** of new checking openings vs. Chase at **9%**

Gen Z: Most AI-Receptive

62% have already used AI for financial planning guidance

Gen Z: Leading Indicator

If AI works for the hardest audience, it extends to everyone else

Customer Research

Gen Z Is in Financial Crisis — The Data

Financial Literacy

Scored **38%** on the P-Fin Index — 11 pts below the national average. Nearly half don't know what affects their credit score.

Credit in Decline

Average FICO: **676** — steepest YoY drop of any generation. **54%** feel anxious checking their credit score.

Overdrafts & Fees

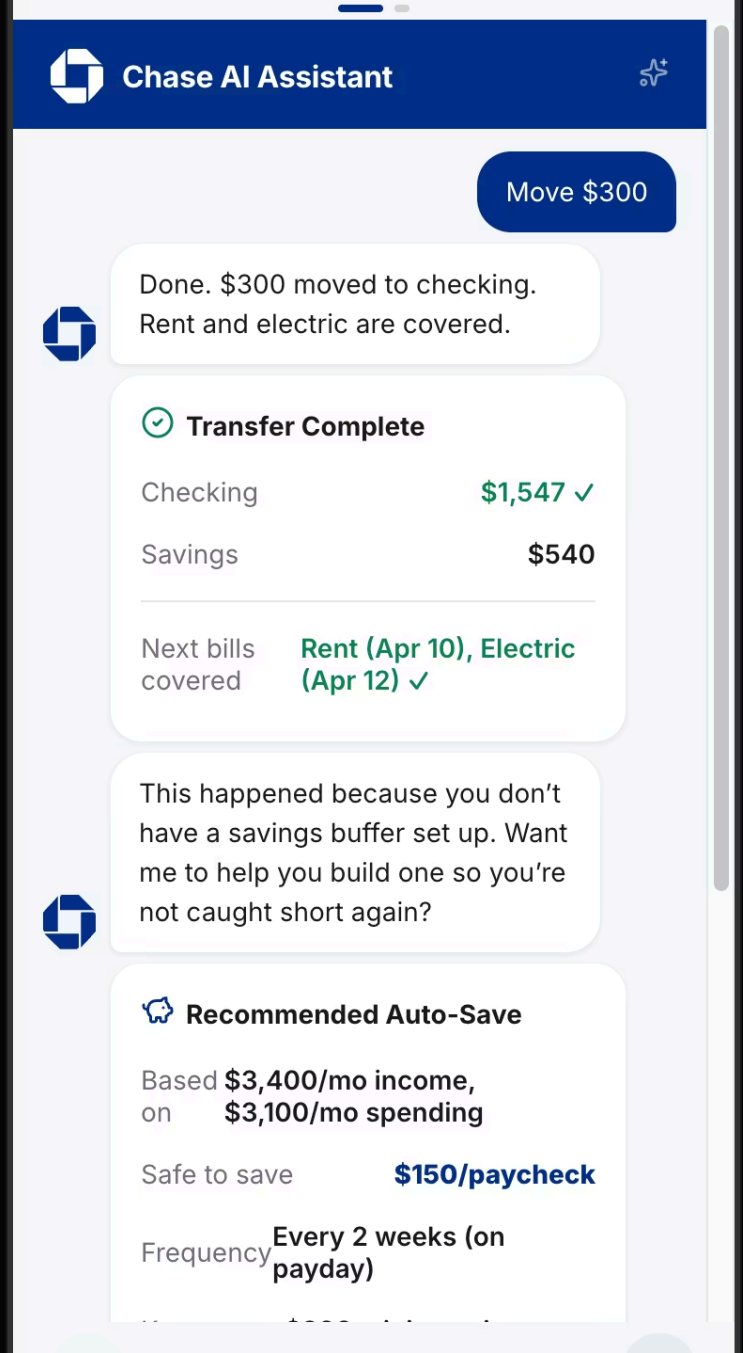
2/3 overdrew their accounts in 2025. Average fee: **$26.61/transaction.** 10%+ overdraft more than 10× a year.

Savings Crisis

Median emergency fund: **$400** — 5× less than boomers. **55%** can't cover 3 months of expenses.

But They Want Help

62% already use AI for finance. **75%** say AI lets them ask questions they'd be too embarrassed to ask a human. **82%** would switch banks for a better digital experience.



Problem Definition

Dashboard, Not Coach

Gen Z checks their balance → sees it declining → doesn't know what to do → **leaves feeling worse.** The app shows data but offers no path forward.

> Only **18%** of users trust AI to make financial decisions autonomously. They want help. They don't want autopilot.

Today

Log in → see balance → log out. Zero guidance, zero action.

The Opportunity

AI that **educates, nudges, and acts** — with explicit user permission at every step.

Market Analysis

The Competitive Landscape — and the White Space

Player	What They Do	What's Missing
BofA Erica	42M users, 3B+ interactions. Command-based: you ask, it answers.	Not agentic. No proactive scanning, suggestions, or actions.
Chime	Zero-fee, **SpotMe, Credit Builder.** Wins Gen Z on simplicity.	No AI story. Simple products, no intelligence layer.
Credit Karma	140M members, deep credit data, GenAI summaries.	Data layer only — can't move money or take banking actions.
Public.com ★	First agentic brokerage. Plain-language intent → agent executes with approval.	Investment-only. Doesn't touch banking, spending, or bills.
ChatGPT / Gemini	Generic financial Q&A, popular for "embarrassing" questions.	No access to your data. Can't act.

The white space: No one has built a permission-based agentic experience for everyday banking — spending, saving, bills, credit. Public proved the model for investing. We apply it to the rest of financial life.

Technical Honesty

What Can an Agent Actually Do?

Before prioritizing, we must be clear about what's genuinely feasible in a banking context.

Action	How It Works	Feasibility
Read account data, explain concepts	Bank read APIs (balances, transactions, credit score)	✅ Standard
Transfer between your own accounts	Scheduled transfer APIs (checking → savings)	✅ Standard
Send money via Zelle	Zelle API, internal bank transfers	✅ Standard
Set up bank bill pay	Bank sends ACH/check to biller (utilities, rent)	✅ Standard
Identify subscriptions	Transaction categorization on recurring charges	✅ Read-only
Cancel subscriptions	Requires auth with provider (Netflix, Spotify, etc.)	❌ Outside bank's control
Block merchant charges	Card-level controls — bank-dependent	⚠️ Partially feasible

Bank-Initiated Payments ✅

Bill pay, transfers, Zelle — the bank sends money. The agent can set these up.

Merchant-Initiated Charges ⚠️

Netflix charges your card — the bank doesn't pay Netflix. The agent can *identify, alert, and flag* — not cancel or pay on your behalf.

THE SOLUTION

Agentic Financial Assistant

Not a chatbot that answers questions — an agent that **educates, recommends, and acts** on your behalf with explicit approval at every step.

1

1. Financial Coach

Judgment-free Q&A personalized to your account. "Your score is 676 — here's what's pulling it down and 3 things you can do this month." *Read-only.*

2

2. Savings Agent

Analyzes income and spending, proposes auto-transfers. "You have $340 left this pay period. Move $100 to savings and keep a $240 buffer?" Uses standard bank transfer APIs.

3

3. Payment Agent

"Send Mom $50" → Zelle. "Pay rent" → bank bill pay. Natural-language payments over existing Zelle and ACH infrastructure — no new payment rails needed.

4

4. Subscription Awareness

Identifies 8 subscriptions totaling $127/mo. "You haven't used Paramount+ in 4 months — $13.99/mo. Here's how to cancel." Agent surfaces; user acts.

5

5. Spending Insights

Category budgets, spending alerts, merchant-level controls where the bank supports them. "You've spent $180 on dining — 20% above usual." Proactive, not reactive.

The Permission Model Is the Product: Every action follows **suggest → explain why → user confirms → execute.** The agent never acts without approval.

Roadmap

Prioritization — Hypotheses & Milestones

Framework: Hypothesis Testing Speed × User Impact × Technical Feasibility. Milestones overlap — build next while validating current.
What's gated is launch, not development.

M1: Coach + Foundation (Start Here)

H1: "Will Gen Z engage with a judgment-free AI coach in their banking app?" Read-only, fastest to ship, maximum learning. Every conversation informs what to build next.

M2: Action Layer

H2: Savings Agent — automated transfers, round-ups, overdraft buffers.
H3: Payment Agent — Zelle + bill pay via natural language. Built on standard APIs, informed by Coach data.

M3: Intelligence Layer

H4: Subscription Awareness (read-only) · **H5:** Spending Insights + merchant controls. Requires rich Coach conversation data to personalize effectively.

Technical Architecture

How It's Built — and What We've Designed Around

Architecture Principles

Zero-PII Model Interaction

LLM never touches raw customer data. Only sanitized summaries and structured fields — based on Wells Fargo's production architecture.

RAG for Financial Grounding

Product info, compliance rules, and account summaries as context. Prevents hallucination on financial facts.

HITL Tiers

Tier 1: Autonomous (Q&A). Tier 2: User confirm (auto-save). Tier 3: Explicit approval (auto-pay). Tier 4: Human review (large transfers).

Limitations — Designed Around, Not Ignored

Hallucination: Zero Tolerance

All advice grounded via RAG and direct user data queries.

The Klarna Lesson

Went AI-only, cut 700 agents, then rehired when quality dropped. Design for human escalation from day 1.

Proxy Discrimination

AI can produce disparate outcomes through zip code, language, and spending patterns — without using protected characteristics. Requires active monitoring.

Go-To-Market

Launch Tight, Loosen with Data

Milestones	What Ships to Users	Agent Tools Required	Eval & Quality
M1: Pilot	Financial Coach (read-only)	`get_account_balance(account_id)` `get_transaction_history(account_id, date_range)` `get_credit_score(user_id)` `explain_concept(topic)` `categorize_spending(transactions)`	Agentic eval on 100% of responses + human benchmarking on sample
M2: Expand	+ Savings Agent, Payment Agent	`create_savings_rule(user_id, trigger, amount)` `schedule_transfer(from_account, to_account, amount, date)` `send_zelle(recipient, amount, memo)` `setup_bill_pay(biller, amount, frequency)`	Eval pipeline validated; reduce human sample rate
M3: Scale	+ Subscription Awareness, Spending Controls	`detect_subscriptions(transactions)` `flag_subscription(merchant_id, user_id)` `set_spending_alert(category, threshold)` `block_merchant(merchant_id, user_id)`	Automated eval with periodic human validation sweeps

Factual Accuracy

Grounded in account data + agentic queries to SOPs. Every response is grounded automatically.

Tone

Judgment-free? Conversational, not corporate? Calibrated by human reviewers and UXR.

Hallucination Detection

Claims not supported by verified data are flagged before they reach users via guardrail hooks.

Measurement & Discussion

Success Metrics

North Star: Weekly Active Agent Users (WAU) — captures real habit formation, not one-time curiosity.

Engagement	Conversations/user/week · Task completion rate · 7d and 30d return rate
Financial Outcomes	Overdraft reduction % · Auto-save adoption · Net savings growth/user
Business Impact	Gen Z 90d/180d retention lift · NPS delta · Cost-to-serve reduction
Guardrails	Eval accuracy vs. human baseline · Hallucination rate · Regulatory flags

Questions & Comments